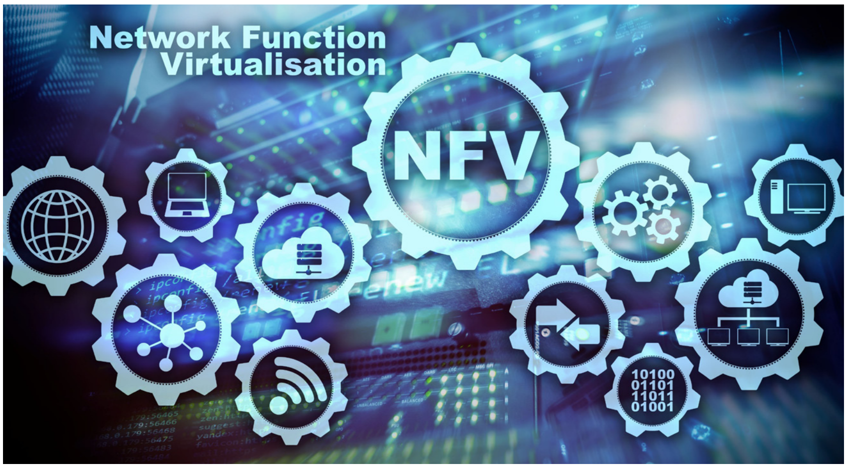
Unlocking the Potential of Virtualised Network Functions and Service Function Chaining on OpenStack

In the first part of this article (carried in the previous issue of OSFY), we discussed the step-by-step procedure of OpenStack installation with Tacker. This second part gives an overview of virtualised network functions and how they can be deployed on the OpenStack platform. It also explores the concept of service function chaining (SFC) and how it can be used to string together multiple VNFs to create a service chain.



As cloud computing continues to gain momentum, the use of virtualised network functions (VNFs) is becoming increasingly popular. With VNFs, software implementations of network functions can be deployed on virtual machines instead of proprietary hardware devices, providing greater flexibility and scalability.

Moving from traditional networking to software defined networking (SDN)

In traditional networking, network devices such as routers and switches are used to forward traffic between different networks and devices. These devices are typically dedicated hardware appliances that are configured with specific rules and settings to control the flow of traffic. Traditional networking architectures rely on a hierarchical structure, with devices at the edge of the network connecting to devices at the core of the network. Traffic is forwarded between these devices using a variety of protocols, such as TCP/IP and Ethernet.

Traditional networking is widely used in a variety of environments, including enterprise networks, service provider networks, and the internet. It is a reliable and well-established technology that has been in use for many years. However, traditional networking can be inflexible and can be difficult to modify or update, particularly in large and complex networks. As a result, newer technologies such as software defined networking (SDN) and network functions virtualisation (NFV) have been developed to enable more flexible and agile networking.

Software defined networking: SDN is an approach to networking that frees administrators from having to manually configure each network device by allowing them to define and manage network policies and behaviours using software. In SDN design, the network is defined and managed by a central controller, while networking components (such as switches and routers) just serve as simple forwarding elements that execute the controller's commands.

One of the main advantages of SDN is that it enables network administrators to respond to fluctuations in network requirements more rapidly and easily. Making modifications to individual devices can be time-consuming and error-prone with standard networking techniques, especially in large